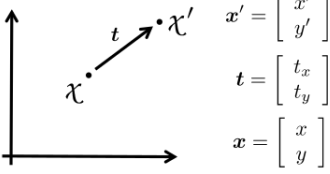
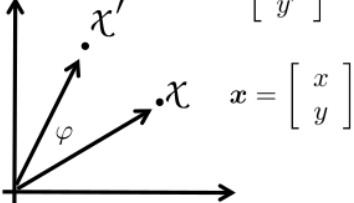
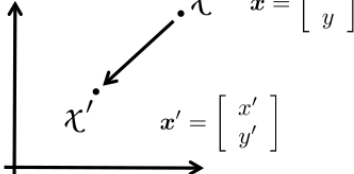


CS: Formulae

Translation	Rotation	Scale Change
 $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ $t = \begin{bmatrix} t_x \\ t_y \end{bmatrix}$ $x = \begin{bmatrix} x \\ y \end{bmatrix}$	 $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$ $x = \begin{bmatrix} x \\ y \end{bmatrix}$	 $x = \begin{bmatrix} x \\ y \end{bmatrix}$ $x' = \begin{bmatrix} x' \\ y' \end{bmatrix}$
$x' = \mathcal{T}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$x' = \mathcal{R}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$	$x' = \mathcal{D}(x) : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix}$
Sum	Matrix Multiplication	Matrix Multiplication

Similarity Transformation

-	In 2D	In 3D
Scale Change	1 DoF	1 DoF
Rotation	1 DoF	3 DoF
Translation	2 DoF	3 DoF
Sum	4 DoF	7 DoF
-	$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix} \begin{bmatrix} \lambda x \\ \lambda y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$	$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_X \\ t_Y \\ t_Z \end{bmatrix}$

Rotation Matrix - Constraints

$$\begin{array}{c}
 c_i^T \cdot c_j = 0 \\
 \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \\
 \begin{array}{ccc} c_1 & c_3 & c_2 \end{array} \\
 |c_i|^2 = 1
 \end{array}$$

3D Rotation Parameterisations

Representation	# Parameters	Unique	Readable	Singularities
Rotation Matrix	9	✓	✗	✗
Euler Angles	3	✗	✓	✓
Rotation Axis and Angle	4(3)	(✗)	(✓)	(✓)
Quaternion	4(3)	✓	✗	✗

Rotation Matrix $\leftrightarrow$ Euler Angles

$$R_A(\alpha, \beta, \gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha) = \begin{bmatrix} \cos \gamma \cos \beta & -\sin \gamma \cos \alpha + \cos \gamma \sin \beta \sin \alpha & \sin \gamma \sin \alpha + \cos \gamma \sin \beta \cos \alpha \\ \sin \gamma \cos \beta & \cos \gamma \cos \alpha + \sin \gamma \sin \beta \sin \alpha & -\cos \gamma \sin \alpha + \sin \gamma \sin \beta \cos \alpha \\ -\sin \beta & \cos \beta \sin \alpha & \cos \beta \cos \alpha \end{bmatrix}$$

If $\cos \beta \neq 0$:

- $\alpha = \text{atan2}(R_{32}, R_{33})$
- $\beta = \text{atan2}(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2})$
- $\gamma = \text{atan2}(R_{21}, R_{11})$

If $\cos \beta = 0$:

$$R_A = \begin{bmatrix} 0 & -\sin(\gamma - \alpha) & \cos(\gamma - \alpha) \\ 0 & \cos(\gamma - \alpha) & \sin(\gamma - \alpha) \\ -1 & 0 & 0 \end{bmatrix}$$

Rotation Matrix to Euler Angles

$$R_A(\alpha, \beta, \gamma) = R_3(\gamma)R_2(\beta)R_1(\alpha) =$$

$$\begin{bmatrix} \cos \gamma \cos \beta & -\sin \gamma \cos \alpha + \cos \gamma \sin \beta \sin \alpha & \sin \gamma \sin \alpha + \cos \gamma \sin \beta \cos \alpha \\ \sin \gamma \cos \beta & \cos \gamma \cos \alpha + \sin \gamma \sin \beta \sin \alpha & -\cos \gamma \sin \alpha + \sin \gamma \sin \beta \cos \alpha \\ -\sin \beta & \cos \beta \sin \alpha & \cos \beta \cos \alpha \end{bmatrix}$$

$\gamma = \text{atan2}(R_{21}, R_{11})$
 $\beta = \text{atan2}(-R_{31}, \sqrt{R_{32}^2 + R_{33}^2})$
 $\alpha = \text{atan2}(R_{32}, R_{33})$

Axis Angle Representation

- $\theta, \mathbf{r} = [r_1, r_2, r_3]^T$ with $\|\mathbf{r}\| = 1$
- Alternative Minimal Representation (3 parameters):
 - $\mathbf{r} = [r_1, r_2, r_3]^T$ with $\|\mathbf{r}\| = \theta$

Axis Angle $\rightarrow$ Rotation Matrix

- Rodrigues Formula:

$$R_{\mathbf{r}, \theta} = \mathbf{I}_3 + \sin \theta \mathbf{S}_{\mathbf{r}} + (1 - \cos \theta) \mathbf{S}_{\mathbf{r}}^2$$

- With $\mathbf{S}_{\mathbf{r}} \doteq \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$
 - Normalised:
 - $R_{\theta} = \mathbf{I}_3 + \frac{\sin |\theta|}{|\theta|} \mathbf{S}_{\theta} + \frac{1 - \cos |\theta|}{|\theta|^2} \mathbf{S}_{\theta}^2$
 - with $\boldsymbol{\theta} = \theta \mathbf{r}$

Rotation Matrix to Axis-Angle

- $\theta = \text{atan2}(|\boldsymbol{\alpha}|, \text{tr} R - 1)$
- with:
 - $\boldsymbol{\alpha} = - \begin{pmatrix} r_{23} - r_{32} \\ r_{31} - r_{13} \\ r_{12} - r_{21} \end{pmatrix} = 2 \sin \theta \mathbf{r}$
 - $\text{tr}(R) = r_{11} + r_{22} + r_{33} = 1 + 2 \cos \theta$
 - $\text{tr}(R)$: Trace - Sum of the main diagonal elements
- 3 cases for axis:

1. $\theta = 0 \leftrightarrow R = I$: Singularity

2. $\theta = \pi$: Sign of Axis is irrelevant:

- $\mathbf{r}\mathbf{r}^T = \frac{R+I}{2}$

- $\mathbf{r}$ given by any column of R

3. Otherwise $R = \frac{a}{|a|}$

Quaternion

- Quaternion $\mathbf{q} = \begin{bmatrix} q \\ \mathbf{q} \end{bmatrix}$ is 4 D Vector with $\mathbf{q} = [q_1, q_2, q_3]^T$

- Can be interpreted as

1. Complex number

- 1D real number q and

- 3D complex part $\mathbf{q} = q + q_1i + q_2j + q_3k$

- with 3 complex units i, j, k

- for which holds $i^2 = j^2 = k^2 = ijk = -1$

2. Tuple

- $\mathbf{q} = (q, \mathbf{q})$

- Follow rules of certain algebra

Addition	Multiplication	Inverse	Rotations
$\mathbf{p} = \mathbf{q} + \mathbf{r}$	$\mathbf{p} = \mathbf{q}\mathbf{r}$	$\mathbf{p} = \mathbf{q}^{-1}$	
$(p, \mathbf{p}) = (q + \mathbf{r}, \mathbf{q} + \mathbf{r})$	$(p, \mathbf{p}) = (q\mathbf{r} - \mathbf{q} \cdot \mathbf{r}, \mathbf{r}\mathbf{q} + \mathbf{q}\mathbf{r} + \mathbf{q} \times \mathbf{r})$	$\mathbf{q}^{-1} = \frac{\mathbf{q}^*}{ \mathbf{q} ^2}$	$\mathbf{r} = [r_1, r_2, r_3]^T$ with $ \mathbf{r} = 1$
$\begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} q_0 + r_0 \\ q_1 + r_1 \\ q_2 + r_2 \\ q_3 + r_3 \end{bmatrix}$	$\begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} q_0r_0 - q_1r_1 - q_2r_2 - q_3r_3 \\ q_1r_0 + q_0r_1 + q_2r_3 - q_3r_2 \\ q_2r_0 + q_0r_2 - q_1r_3 + q_3r_1 \\ q_3r_0 + q_0r_3 + q_1r_2 - q_2r_1 \end{bmatrix}$	$\begin{bmatrix} q \\ -\mathbf{q} \end{bmatrix} \frac{1}{q_0^2 + q_1^2 + q_2^2 + q_3^2}$	$\mathbf{q} = \begin{bmatrix} q \\ \mathbf{q} \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \sin(\theta/2)\mathbf{r} \end{bmatrix}$

Quaternion Multiplication

$$\begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} q_0 r_0 - q_1 r_1 - q_2 r_2 - q_3 r_3 \\ q_1 r_0 + q_0 r_1 + q_2 r_3 - q_3 r_2 \\ q_2 r_0 + q_0 r_2 - q_1 r_3 + q_3 r_1 \\ q_3 r_0 + q_0 r_3 + q_1 r_2 - q_2 r_1 \end{bmatrix}$$

Quaternion and Rotation Matrix